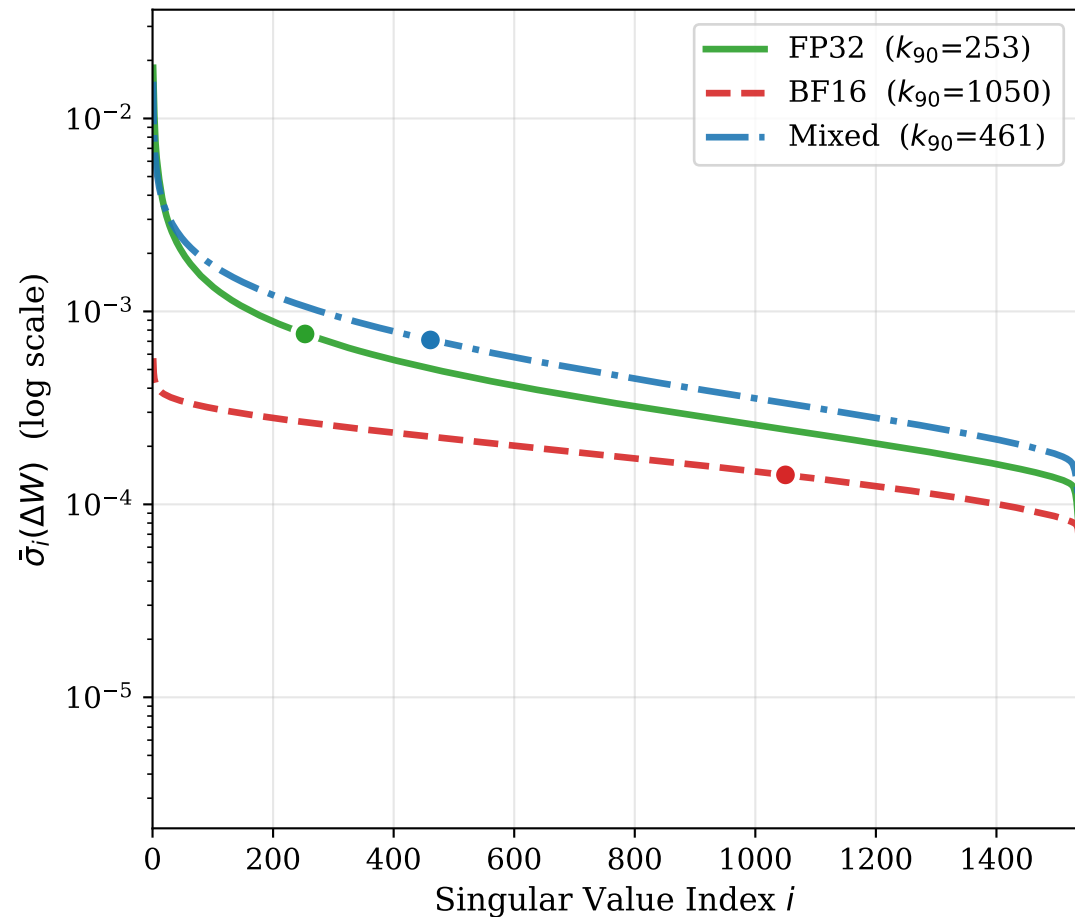


**Spectral Decay of  $\bar{\sigma}_i(\Delta W)$  with  $k_{90}$  Marked**  
( $\bar{\sigma}_i$  and  $k_{90}$  computed from the mean spectrum across all layers/components)

**AdamW + GRPO**



**Muon + GRPO**

